ONLINE CLASSIFIELDS MARKETPLACE

A secure and reliable peer-to-peer (P2P) online marketplace

CS23531 - MINI-PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled CLASSIMARKET- SECURE AND RELIABLE P2P MARKETPLACE is the bonafide work of **PONNILA JUDY A (241001163)**, who carried out the work during the Mini-Project under supervision. Certified further that, based on the task given by the training organisation, the project was completed under their supervision.

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ABSTRACT

The Online Classifieds Marketplace, branded as ClassiMarket, is a web-based platform designed to simplify the buying and selling of used and new products within college campuses, residential communities and nearby localities. The application provides a common digital space where registered sellers can publish product advertisements containing descriptions, prices, categories, locations and images. Buyers can browse advertisements, search using keywords, apply category and price filters, and view seller contact information.

The system is developed using HTML5, CSS3, Bootstrap 5, JavaScript, PHP and MariaDB. PHP handles server-side processing and application logic, while MariaDB stores user, category and listing information. PHP Data Objects (PDO) is used for database connectivity and prepared statements. The application includes user registration and login, listing creation and management, product browsing, search and filtering, and a seller dashboard.

Security measures described in the implementation include password hashing, prepared SQL statements, output escaping, CSRF token validation, session management, image upload validation and listing ownership checks. The project demonstrates the integration of frontend development, backend programming and relational database management in a practical community-oriented marketplace.

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1. INTRODUCTION

1.1 Introduction

Online marketplaces have become an important part of everyday commerce. However, students and members of local communities often rely on informal social media groups or personal contacts to buy and sell used products. These methods may not provide organized categories, structured search, listing status management or convenient product discovery.

ClassiMarket is designed as a simple online classifieds platform that connects buyers and sellers directly. Sellers can publish advertisements for products such as textbooks, electronics, furniture, mobile phones, fashion items and vehicles. Buyers can browse the catalogue, search for products and contact sellers using the information provided in the listing.

1.2 Project Overview

ClassiMarket follows a client-server model. The browser displays the user interface, Apache receives HTTP requests, PHP processes application logic, and MariaDB stores persistent information. The application is organized into modules for authentication, listings, dashboard functions, reusable components and public assets.

1.3 Scope of the Project

- Provide user registration, login and logout facilities.
- Allow authenticated sellers to create, edit, update and delete their own advertisements.
- Allow visitors and buyers to browse product advertisements.
- Provide keyword search, category filtering, price filtering and sorting.
- Display product information and seller contact options.
- Provide sellers with a dashboard showing listing statistics.
- Apply common web security practices to important user and database operations.

1.4 Intended Users

The intended users include college students, faculty members, hostel residents and people in local communities who want to advertise or discover products. The application is suitable for community-level peer-to-peer classified listings.

2. PROBLEM DEFINITION AND OBJECTIVES

2.1 Problem Statement

College campuses and residential communities may not have a dedicated and organized platform for trading used goods. Social media groups can make it difficult to locate specific products because posts are not consistently categorized and may become difficult to search. Listings may remain visible even after an item is sold, and buyers may have limited structured information about the product or seller.

2.2 Proposed Solution

ClassiMarket addresses these issues by providing a centralized classifieds website. It organizes advertisements by category and supports keyword and price-based filtering. Sellers can manage the lifecycle of their listings, while buyers can view product details and contact sellers directly.

2.3 Objectives

- Develop a responsive and user-friendly classifieds website.
- Implement secure user registration and authentication.
- Provide facilities to publish and manage product advertisements.
- Store users, categories and listings in a relational database.
- Implement search, filtering and sorting for product discovery.
- Provide a seller dashboard with summary information.
- Use server-side validation and common web security protections.

2.4 Advantages

- Centralized product advertisements in one application.
- Structured categories for easier browsing.
- Search and filtering reduce the effort required to find products.
- Direct buyer-to-seller communication without a transaction intermediary.
- Seller-side listing management improves listing accuracy.
- Responsive interface supports different screen sizes.

3. SYSTEM ANALYSIS AND REQUIREMENTS

3.1 Existing System

In an informal buying and selling process, users commonly post product details in social media groups or communicate through personal networks. Such methods may lack consistent listing fields, dedicated search filters, seller dashboards and clear listing status management.

3.2 Limitations of the Existing System

- Product information may be scattered across multiple posts.
- Search and category filtering may be limited.
- Sold items may continue to appear in old posts.
- Seller listing management is not centralized.
- Product information may be incomplete or inconsistent.

3.3 Proposed System

The proposed system is a browser-based classifieds marketplace. It stores advertisements in a relational database and provides a structured interface for browsing and managing them. Users can search by keyword, category and price range. Sellers can manage their own advertisements through authenticated pages.

3.4 Functional Requirements

- User registration with input validation.
- Login and logout with session handling.
- Creation of product advertisements with image upload.
- Viewing of advertisement details.
- Editing, status updating and deletion of seller-owned advertisements.
- Browsing, searching, filtering and sorting of advertisements.
- Dashboard metrics for the authenticated seller.

3.5 Non-Functional Requirements

- Usability: clear navigation and understandable forms.
- Performance: indexed database fields and appropriately filtered queries.
- Security: prepared statements, password hashing, CSRF validation and output escaping.
- Maintainability: modular PHP files and reusable functions.
- Portability: execution in a compatible PHP and MariaDB environment.
- Responsiveness: Bootstrap-based layouts for desktop and smaller screens.

3.6 Hardware and Software Requirements

Category	Requirement
Hardware	Computer or laptop capable of running a local web development environment
Operating System	Windows (development environment described)
Web Server	Apache HTTP Server
Backend	PHP 8.2.12
Database	MariaDB 10.4.32
Frontend	HTML5, CSS3, Bootstrap 5.3.3, JavaScript
Development Environment	XAMPP and Visual Studio Code
Browser	Modern web browser

4. SYSTEM DESIGN AND ARCHITECTURE

4.1 Architecture

ClassiMarket uses a lightweight procedural PHP architecture. The browser sends HTTP requests to Apache. PHP pages and reusable helper functions process the requests, validate input, perform authentication and communicate with MariaDB through PDO. The response is returned to the browser as HTML.

Layer	Responsibility
Presentation Layer	HTML, CSS, Bootstrap and JavaScript render pages and handle client-side interactions.
Application Layer	PHP processes authentication, listing operations, search and validation.
Data Access Layer	PDO provides prepared database operations and exception handling.
Database Layer	MariaDB stores users, categories and listings.
Web Server	Apache serves the application and executes PHP requests.

4.2 Application Data Flow

- User opens a page or submits a form in the browser.
- Apache receives the HTTP request and routes it to the relevant PHP page.
- PHP checks the session, validates input and applies authorization rules where required.
- PDO executes the relevant prepared SQL statement against MariaDB.
- The application processes the returned data and generates an HTML response.
- The browser displays the result to the user.

4.3 Directory Structure

The project is organized into folders according to responsibility. The following structure is based on the project layout supplied for this report:

```
online-classifieds/
index.php
 README.md
              .gitignore
config/database.php
includes/header.php
includes/footer.php
includes/functions.php
auth/register.php
auth/login.php
auth/logout.php
listings/index.php
listings/view.php
listings/create.php
listings/edit.php
listings/delete.php
dashboard/index.php
assets/css/style.css
assets/js/main.js
database/schema.sql
database/seed.sql
uploads/listings/
```

4.4 Main Modules

Module	Purpose
Authentication	Registration, login, logout and session handling.
Browse and Search	Displays advertisements and supports keyword/category/price filters.
Listing Management	Creates, views, edits, updates and deletes advertisements.
Seller Dashboard	Displays listing counts and active inventory value.
Common Components	Reusable navigation, footer, helper and security functions.
Database	Stores user, category and advertisement records.

5. TECHNOLOGY STACK

Technology	Purpose in ClassiMarket
HTML5	Defines page structure, forms, navigation and content sections.
CSS3	Controls colours, typography, spacing, layout and responsive presentation.
Bootstrap 5.3.3	Provides responsive grid layouts and ready-made interface components.
Bootstrap Icons 1.11.3	Provides icons for navigation, categories and actions.
JavaScript ES6+	Supports image preview, client-side checks and interface interactions.
PHP 8.2.12	Implements server-side logic, authentication and listing operations.
MariaDB 10.4.32	Stores users, categories and listings using relational tables.
PDO	Connects PHP to the database and supports parameterized queries.
Apache 2.4.58	Receives HTTP requests and serves the PHP application.
XAMPP	Provides the local Apache, PHP and MariaDB development environment.
Visual Studio Code	Used to edit and manage project source files.

5.1 Development Environment

The application is hosted locally using XAMPP. Apache is configured on port 80. The XAMPP MariaDB service is configured on port 3308 because a separate MySQL service on the computer uses the default port 3306. The database connection settings and phpMyAdmin configuration must use the same MariaDB port.

5.2 Database Connectivity

The application uses PHP Data Objects (PDO) to connect to MariaDB. Prepared statements are used for queries containing user-provided values. PDO exception mode is used to support consistent database error handling.

6. DATABASE DESIGN

6.1 Database Overview

The database is named online_classifieds. It stores account information, product categories and advertisements. The InnoDB storage engine is used to support transactions and foreign key constraints.

6.2 Users Table

The users table stores account information required for registration and authentication. The email field is unique so that the same email address cannot be used for multiple accounts. Passwords are stored as hashes rather than plaintext.

Field	Purpose
id	Primary key identifying each user.
name	Name of the registered user.
email	Unique email address used for login.
password	Stored password hash.
created_at	Account creation timestamp.

6.3 Categories Table

The categories table stores the available product categories. Category names and slugs are used to identify and filter listings.

Field	Purpose
id	Primary key identifying a category.
name	Display name of the category.
slug	Unique URL-friendly category identifier.

6.4 Listings Table

The listings table stores product advertisements and connects each advertisement to its seller and category.

Field	Purpose
id	Primary key for a listing.
user_id	Foreign key referencing the seller in users.
category_id	Foreign key referencing categories.
title	Product title.
description	Product description.
price	Product price stored as a decimal value.

Field	Purpose
location	Location associated with the listing.
image	Stored image filename or path.
status	Listing state such as active, sold or archived.
created_at / updated_at	Creation and last-update timestamps.

6.5 Relationships and Constraints

• Each listing is associated with a user through listings.user_id.

• Each listing belongs to a category through listings.category_id.

• The user relationship uses ON DELETE CASCADE as described in the project design.

• The category relationship uses ON DELETE RESTRICT to prevent removal of a category that is still referenced.

• Indexes are used on important listing fields such as category, user and status to support common queries.

6.6 Entity Relationship Description

The database contains three principal entities: users, categories and listings. One user can create multiple listings, and one category can contain multiple listings. Each listing references exactly one seller and one category through its foreign keys.

7. IMPLEMENTATION

7.1 Homepage

The homepage is the main entry point of ClassiMarket. It contains the brand name, navigation links, a hero section, a search bar, category cards and recent listings. The supplied interface shows the headline “Buy and Sell Anything, Fast & Easy” and category cards for Books, Electronics, Fashion, Furniture, Mobile Phones and Vehicles.

7.2 Registration and Login

The registration module accepts user details and validates them on the server. It checks email uniqueness and stores a password hash using PHP password hashing. The login module retrieves the account using the supplied email and verifies the password using password_verify(). Session identifiers are regenerated after successful authentication.

7.3 Product Browsing and Search

The Browse Ads page retrieves advertisements from the database. Keyword search can match product titles, descriptions and locations. Category and minimum/maximum price conditions can be combined to narrow the results. Sorting options described for the application include newest, oldest, price low-to-high and price high-to-low.

7.4 Creating an Advertisement

Authenticated sellers can submit a product title, description, category, price, location and image. The server validates the submitted data and checks the uploaded file. Valid images are stored in the listings upload directory using randomized filenames, and the listing information is stored in MariaDB.

7.5 Viewing Product Details

The product details page displays the selected advertisement, including its image, title, description, price, category and location. Seller contact options may be presented through telephone and email links where the seller has provided the corresponding information.

7.6 Editing and Deleting Advertisements

The edit module allows a seller to update an advertisement they own. The seller can modify listing information and update its status, for example to sold or archived. The delete module uses a POST request and CSRF verification. Ownership is checked before a record is modified or deleted.

7.7 Seller Dashboard

The seller dashboard provides a summary of the user's advertisements. The described metrics include total advertisements, active listings, sold listings and the gross value of active inventory. A management table provides access to listing actions.

7.8 Reusable Components

The header and footer are maintained in reusable include files. The common functions file contains helper functions related to authentication, output escaping, CSRF handling and image uploads. This modular structure helps maintain consistency and reduces duplicated code.

7.9 JavaScript Interactions

The JavaScript file supports client-side interactions such as image preview, preliminary file-size checks, delete confirmation prompts and automatic dismissal of alert messages. Server-side validation remains necessary because client-side checks can be bypassed.

7.10 Security Features

Risk	Protection described in the project
SQL Injection	PDO prepared statements and parameter binding.
Cross-Site Scripting	Escaping dynamic output with htmlspecialchars through a reusable helper.
CSRF	Session-based random token and hash_equals verification.
Password Exposure	Password hashing and password_verify during login.
Session Fixation	Session ID regeneration after authentication.
Unsafe Image Upload	File-size, MIME-type and extension validation; randomized filenames.
Unauthorized Listing Changes	Server-side ownership checks for edit and delete operations.

8. RESULTS AND DISCUSSION

8.1 Application Output

The completed interface provides a homepage with navigation, search, category browsing and recent advertisements. The application is intended to support the full marketplace workflow from account creation to listing management.

8.2 Homepage Interface

The homepage presents the ClassiMarket identity in a blue navigation bar. The central hero area includes a search input, category selector and search button. Below it, category cards provide entry points to common product groups. The supplied homepage screenshot can be inserted as Figure 8.1 in the final document.

8.3 Browse Ads

The browsing page presents advertisements and provides search and filtering options. A screenshot of the running Browse Ads page should be inserted as Figure 8.2.

8.4 Authentication Pages

The registration and login pages allow users to create accounts and access protected seller functionality. Insert a screenshot of the actual pages as Figure 8.3.

8.5 Listing Details and Creation

The listing detail page presents product information and seller contact options. The create advertisement page allows an authenticated seller to submit product details and an image. Insert actual screenshots as Figures 8.4 and 8.5.

8.6 Seller Dashboard

The seller dashboard presents summary metrics and a listing management table. Insert a screenshot of the actua dashboard as Figure 8.6.

8.7 Discussion

The project demonstrates how frontend components, server-side PHP processing and relational database operations can be combined in a single web application. Search and filtering support product discovery, while the seller dashboard and listing lifecycle operations support advertisement management. The implemented security checks address common risks associated with user accounts, database queries and file uploads.

9. TESTING

9.1 Testing Approach

Testing is used to check whether the application's individual modules and integrated workflows behave as expected. The project description identifies unit, integration, HTTP authentication and CRUD testing as relevant testing activities. The final report should record only test cases that were actually executed.

9.2 Functional Test Cases

ID	Test Scenario	Expected Result
TC01	Open homepage	Homepage is displayed.
TC02	Open Browse Ads	Advertisement catalogue is displayed.
TC03	Register with valid details	Account is created.
TC04	Register with an existing email	Duplicate registration is rejected.
TC05	Login with valid credentials	User is authenticated.
TC06	Login with invalid credentials	Authentication is rejected.
TC07	Create an advertisement	Listing is stored in the database.
TC08	Search by keyword	Matching listings are displayed.
TC09	Filter by category	Listings from the selected category are displayed.
TC10	Filter by price	Listings within the selected range are displayed.
TC11	Edit own listing	Listing details are updated.
TC12	Delete own listing	Listing is removed.
TC13	Modify another user's listing	Unauthorized operation is rejected.
TC14	Logout	Authenticated session is terminated.

9.3 Security Test Scenarios

Scenario	Expected Result
SQL injection-style input	Input is treated as data by parameterized queries.
Invalid CSRF token	State-changing request is rejected.
Unauthorized listing edit/delete	Ownership check prevents the operation.
Unsupported image file	Upload is rejected.
Image larger than configured limit	Upload is rejected.
Unauthenticated access to protected page	Access is restricted or redirected.

9.4 CRUD Testing

Operation	Description
Create	Insert a new product advertisement.
Read	Retrieve and display advertisements.
Update	Modify advertisement details or status.
Delete	Remove an advertisement.

9.5 Test Result Recording

For the final submission, mark each test case as PASS or FAIL only after executing it in the local application. Where a test was not executed, record it as NOT TESTED. This avoids reporting unverified results.

10. CHALLENGES FACED

10.1 Database Port Conflict

The computer already had a MySQL service using port 3306. To avoid a port conflict, the XAMPP MariaDB service was configured to use port 3308. The project's database configuration and phpMyAdmin settings were aligned with this port.

10.2 Database Connectivity

Correct host, port, database name and credentials were required to establish the PHP-to-MariaDB connection. PDO was used to centralize database access.

10.3 Authentication and Session Handling

Registration and login required validation, password hashing and session management. Session ID regeneration was included to reduce session fixation risk.

10.4 Image Upload Handling

Image upload required checks on file size, MIME type and extension. Randomized filenames were used rather than trusting the original filename.

10.5 Dynamic Search

Combining keyword, category and price filters required conditional query construction. Parameter binding was used for user-provided values.

10.6 Listing Ownership

The application needed to ensure that a seller could modify or delete only their own advertisements. Ownership checks were therefore applied on the server side.

10.7 Responsive Design

Bootstrap components and custom CSS were used to maintain a consistent interface across different screen sizes.

11. CONCLUSION AND FUTURE SCOPE

11.1 Conclusion

ClassiMarket was developed as a community-oriented online classifieds platform for publishing, browsing and managing product advertisements. The project combines HTML, CSS, Bootstrap and JavaScript on the frontend with PHP and MariaDB on the backend.

The main features include user registration and login, product advertisement creation, product browsing, keyword search, category and price filtering, listing modification and deletion, and a seller dashboard. PDO-based database access and security mechanisms such as password hashing, prepared statements, CSRF verification, output

escaping and ownership checks are included in the design.

The project provides practical experience in full-stack web development, relational database design, server-side programming, user authentication and web application security.

11.2 Future Scope

• Real-time buyer-seller messaging using WebSockets.

• Online payment gateway integration.

• Google OAuth or institutional Single Sign-On.

• Location-based search using map services.

• Push notifications for buyer inquiries and listing updates.

• Product ratings and seller reviews.

• Advanced filters for product condition and posting date.

• Cloud deployment for wider availability.

• Administrative dashboard for user and listing moderation.

• Dedicated mobile application.

12. REFERENCES

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• MariaDB Documentation. https://mariadb.org/documentation/

• MySQL Reference Manual. https://dev.mysql.com/doc/

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• Bootstrap Icons. https://icons.getbootstrap.com/

• MDN Web Docs – HTML, CSS and JavaScript. https://developer.mozilla.org/

• PHP PDO Documentation. https://www.php.net/manual/en/book.pdo.php

• OWASP Foundation – OWASP Top 10. https://owasp.org/www-project-top-ten/

• Apache HTTP Server Documentation. https://httpd.apache.org/docs/

• Apache Friends – XAMPP. https://www.apachefriends.org/

APPENDIX A – USER MANUAL

A.1 Starting the Application

• Install and configure XAMPP on the development computer.

• Copy the project folder into the XAMPP htdocs directory.

• Start Apache from the XAMPP Control Panel.

• Start the configured MariaDB service on port 3308.

• Open phpMyAdmin and create or import the online_classifieds database using the supplied SQL files.

• Check the database settings in config/database.php.

• Open http://localhost/online-classifieds/ in a browser.

A.2 Registering and Logging In

Open the Register page and enter the required details. After successful registration, use the registered email address and password on the Log In page. The application verifies the credentials and establishes a session.

A.3 Browsing Advertisements

Open Browse Ads to view available products. Enter a keyword, choose a category or apply price limits as required. Select an advertisement to view its full details.

A.4 Publishing an Advertisement

Log in, select Post Ad, enter the product information, choose a category, provide the price and location, select an image and submit the form. The application validates the input and stores the advertisement.

A.5 Managing Advertisements

Open the seller dashboard to view personal advertisements. Use the available controls to edit details, update the listing status or delete an advertisement.

A.6 Logging Out

Use the Logout option to end the authenticated session.

APPENDIX B – GLOSSARY

Term	Meaning
C2C	Consumer-to-Consumer
CRUD	Create, Read, Update and Delete
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
PHP	PHP: Hypertext Preprocessor
SQL	Structured Query Language
MariaDB	Relational database management system
PDO	PHP Data Objects
HTTP	Hypertext Transfer Protocol
XSS	Cross-Site Scripting
CSRF	Cross-Site Request Forgery
MIME	Multipurpose Internet Mail Extensions
Primary Key	Unique identifier for a database record
Foreign Key	Constraint referencing a record in another table
Prepared Statement	SQL statement using bound parameters
Session	Server-side mechanism for maintaining user state